

Forecasting University System of Maryland Enrollment Rates

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Abstract—The University System of Maryland's enrollment rates are an important indication of the state of the State of Maryland's higher education system, and predicting them is similarly valuable. Both the state of Maryland and the University System of Maryland have groups that create annual forecasts of these enrollment rates using different models. These projections are currently derived from a variety of models, including linear regression. Time series models, particularly ARIMA models, are another common method of forecasting time dependent data. Models were created for undergraduate and graduate enrollment data at each university within the University System of Maryland to produce a forecast which can be compared to existing models. These forecasts are comparable to existing models, but capture different trends based solely on the data without the use of outside information. The model with the best performance will not be known until years into the future.

Index Terms—Enrollment rates, university enrollment, time series, forecasting, ARIMA.

I. INTRODUCTION

ENROLLMENT rates for post-secondary education, and their forecasts, are of interest to a great number of people and organizations, including the University System of Maryland (USM). Enrollment rates have changed significantly over the past few decades. The rise of the internet since 2000 and the acute effects of COVID-19 in 2020 have both impacted university attendance, as will be clearly visualized later. While professional forecasts for USM exist from both the state of Maryland [1] and from USM itself [2], their methods are either a complex model based on regression or unspecified.

This analysis used time series modeling as an alternative method for forecasting to contribute to the existing literature. Reasonably similar results were achieved, but by using a method that may help reveal currently unconsidered trends and patterns.

II. LITERATURE REVIEW

A. Professional Forecasts

Firstly, it is important to understand the existing professional forecasts.

The work done by the University System of Maryland (USM) does not specify the method used, stating instead that projections come from internal campus teams [2]. It may be that different campuses use different methods, or that methods used internally in the system are simply kept private. This paper is primarily focused on the effect of different strategies employed across USM.

The state of Maryland's report, from the Maryland Higher Education Commission (MHEC), contains an overview of the

method used. To summarize a model with a dozen steps: the past ten years were used to estimate population trends both in USM and in Maryland, as well as economic information. A linear regression model was constructed to predict enrollment based on state population, which was then used to model each subcategory of enrollment [1]. This report depends on observed trends about Maryland and higher education, such as a decreasing proportion of the population being enrolled. The Commission used these to create a very specific model, intended to capture the current state of the population of Maryland.

B. Time Series

A time series typically takes a single variable over consecutive, evenly spaced points in time. A specific time series will often have some well understood characteristics.

1) *ARIMA*: The method that this paper uses to model the time series is a very common method called ARIMA, standing for *Autoregressive Integrated Moving Average* [3]–[5]. This model is a generalization of an Autoregressive Moving Average (ARMA) model.

An *autoregressive* (AR) model of order p is used to model data x_t where a given point depends on the past p points, with a parameter ϕ_i for each lag, $i = 1, \dots, p$. Formulaically,

$$x_t = \phi_1 x_{t-1} + \dots + \phi_p x_{t-p} + \epsilon_t,$$

where ϵ_t is a normally distributed error term.

Moving average (MA) modeling of order q is useful when a given point depends on the error terms of the past q points. Or,

$$x_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q},$$

where each ϵ_i is normally distributed and each θ_i is a parameter.

Combining these two models yields an ARMA model of order (p, q) , which is very powerful but only on data satisfying specific conditions. In particular, the data is required to be *stationary*.

2) *Stationarity*: Stationary data are data that have a constant mean and constant variance. For example, sales at a successful company trend upwards over the lifetime of the company, so the mean increases over time. If the data are not stationary, then the ARMA model will not necessarily be accurate.

In order to transform non-stationary data into stationary data, the most common procedure is to take a difference of order d – this is the *integrating* of the ARIMA model. An order d difference on a time series $\{x_t\}_{t \geq 0}$ corresponds to

$$x_t^{(1)} = x_t - x_{t-1}; \quad x_t^{(d)} = x_t^{(d-1)} - x_{t-1}^{(d-1)}.$$

If there is a constant trend in the series, then subtracting consecutive points will result in the trend term of both points canceling out.

In order to assess stationarity, the following are three standard methods or tests. Firstly, a visual assessment is highly important. It can be very easy to determine if data have a non-constant mean – is there a noticeable trend or does the mean appear piecewise – or if the data have non-constant variance – clear segments of the data where variation is noticeably different than the rest of the data. However, there are cases where it is not so easy to tell, and a statistical test is desired to give more concrete evidence for stationarity.

One common test that can be applied is an Augmented Dickey-Fuller (ADF) test. This estimates whether a time series has a unit root, where the null hypothesis is that the data do have a unit root, and are thus non-stationary [6]. A second common test is a Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test, which can be used to determine if a time series is stationary around a deterministic trend [7]. Note that the KPSS test differs from the ADF test in two important ways: the null hypothesis is stationarity in KPSS, not the opposite; and the test allows for a trend to exist, whereas ADF requires the mean to be zero.

It is important to note that stationarity is not a clear-cut binary fact about data. Sometimes, it may be worth considering an order of differencing that only one test says is non-stationary. Using this differencing may create a simpler model, sometimes with a lower AIC than a more complex model.

3) *Determining ARIMA model order:* The order of differencing applied to the data to transform it to stationary data is the d in the triplet of ARIMA order (p, d, q) . Determining p and q , however, is an even less precise science. To start, the autocorrelation function (ACF) and partial autocorrelation function (PACF) are applied to the differenced data plotted. These functions give an idea of the relation between data points and previous data points, which is clearly highly related to both the AR and MA models. The correlation of each term x_t with the observation at lag k , or x_{t-k} , is calculated by $ACF(k) = \frac{\text{Cov}(x_t, x_{t-k})}{\sqrt{\text{Var}(x_t) \text{Var}(x_{t-k})}}$. The PACF works similarly, but attempts to remove the effect of intermediate lags by conditioning all the above terms on those intermediate values, $x_{t-1}, \dots, x_{t-k+1}$.

On stationary data, the PACF and ACF plots will each have a small number, perhaps zero, of significant, nonzero values, followed by many non-significant values at the remaining lags, with perhaps one or two significant values at much higher lags. For non-seasonal data such as enrollment rates, the number of spikes at the beginning of the plot is most important. On most realistic data, this is between zero and three, with three being uncommon. The number of spikes in the PACF graph is usually the desired order of the AR model, p , and the ACF graph corresponds roughly to the total order of both the AR and the MA component, $p + q$.

Unfortunately, determining the best-fitting order for all three components can be difficult. In particular, sometimes the process of manually fitting (p, d, q) will result in a model with a certain AIC, but there exists another model with an improved

AIC that could not be derived from the plots. However, the function `auto.arima()` from the package `forecast` [8] employs an algorithm to search through reasonable values for p, d and q , and returns the model with the best AIC (or similar criterion by user specification). This method can often lead to better results, favoring simpler models, but may not be the ideal solution to a specific situation.

One final note about ARIMA modeling is that for $d \leq 1$, it is possible to have a non-zero mean, typically referred to as a *mean* if $d = 0$ and *drift* if $d = 1$. This allows the model to increase by a constant at each level, capturing trends that the AR and MA models cannot sufficiently explain. This value is simply added to the right hand side of the full ARIMA model.

Thus, the final model ARIMA model can be written as

$$x_t^{(d)} = \mu + \phi_1 x_{t-1}^{(d)} + \dots + \phi_p x_{t-p}^{(d)} + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}.$$

4) *Model Diagnostics:* Once a model has been created, it should be checked for quality of fit. The most important way to do so is to investigate the model's residuals, calculated as the difference between the actual value and the predicted value. Like regression models, the residuals approximate the error term of the model, which is normally distributed. As such, if the model's residuals are not normally distributed, there is evidence of a poor fit.

For an ARIMA model, however, it is important that there is no significant autocorrelation between different lags of residuals, as this may indicate the residuals are not independent, which is another model assumption. This property is easily examined with a Ljung-Box test, which computes a modified sum of a number of lags, which approximates a chi-squared distribution. The null hypothesis is that the data are uncorrelated. As such, an ideal result is a non-significant p -value.

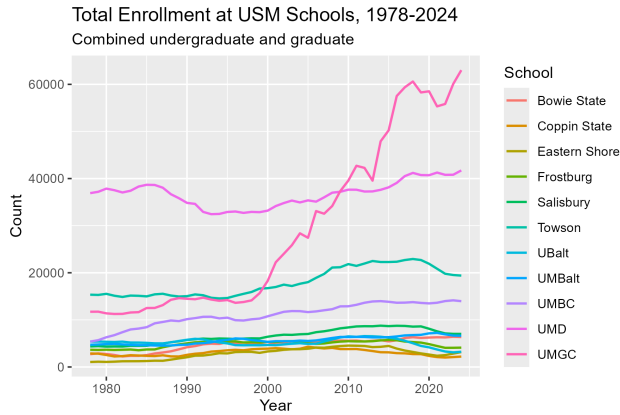
Conveniently, the Ljung-Box test and visual assessments of the distribution of residuals for normality are quickly computed with the `check.residuals()` function from `forecast` [8].

III. DATA

A. Raw Data

The primary data source is past USM enrollment rates, available through the Institution Data Dashboard, called IRIS [9]. The website primarily functions as a rich source for current USM statistics, but also contains historical enrollment data from 1978 to 2024, provided by the USM Institutional Research Information System. This includes system-wide total enrollment and per-university enrollment, both split by under-

graduate and graduate as well as totals.



Upon visual investigation, there are a few notable pieces of information.

Firstly, there is a major outlier in trend in the University of Maryland, Global Campus. Around 2000, when the internet started to become more prevalent, the online-focused university began its explosion in popularity from an average size to the largest university within USM.

Next, there are three main groups of universities within USM. University of Maryland, Global Campus, as discussed previously, is an important outlier. Then, there is the University of Maryland, College Park; Towson University; and University of Maryland, Baltimore County, which have notably higher enrollment and similar trends. And, there are the remaining universities, which all have similar trends and total counts of enrollment.

Finally, there is a dip in enrollment around 2020 across almost all universities, with Towson University being particularly notable. This lines up with expectations around the COVID-19 pandemic and its effect on society and enrollment. However, many of the affected universities have still not recovered from this dip, particularly those with already lower enrollment.

B. Forecasts

As mentioned previously, two sources of existing professional forecasts come from the Maryland Higher Education Commission [1] and the University System of Maryland [2]. Both of these are ten-year projections for 2024-2033 but with differing results. These split students by undergraduate and graduate, as well as full-time and part-time within these categories.

There are two important facets to this data to note. Firstly, as will be visually highlighted when performing comparisons in "Results", the MHEC model has much higher numbers in general. Given how smooth the forecasts are, it is likely the case that this is the result of a subtle difference in how enrollment is calculated, such as counting part-time students more generously.

Also, these forecasts were conducted after the 2023-2024 school year was completed. So, their ten-year forecasts consist of predictions for 2024-2033 fall enrollment. However, the data used for the time series model included fall enrollment

of 2024 as well, due to the time this project began. As such, only the nine year range of 2025-2033 will be considered.

IV. METHODOLOGY

The original forecasts this analysis plans to compare to feature estimates for student enrollments broken down by undergraduate/graduate as well as full-time/part-time at the least. However, the historical enrollment rates do not split by full-time/part-time. As such, the forecasted estimates will be combined to get two values, one for undergraduate and one for graduate, per university per year.

Also note that the original projections only give numbers, and do not provide confidence intervals. To match format and to reduce clutter of data, this report omits confidence intervals from reporting.

The following analysis was conducted in R [10]. Key packages of use were

- `tidyverse` for data manipulation and visualization [11]
- `tseries` for stationarity tests via `adf.test()` and `kpss.test()` [12]
- `forecast` for constructing ARIMA models via `Arima()` and `auto.arima()`, performing diagnostics with `check.residuals()`, and using `forecast()` to easily generate time series forecasts [8].

In order to generate forecasts, two models were created for each university for a total of twenty-two models. For each time series, a common procedure was followed:

- 1) Check data for stationarity via visual inspection, ADF test, and KPSS test
- 2) If data is non-stationary, increase the order of differencing by one and repeat step 1)
- 3) Once data is stationary, graph ACF and PACF of the differenced time series and estimate ARIMA order
 - Note: If data visually appears stationary and only one test disagrees, estimate ARIMA order for this order of differencing as well
- 4) Create ARIMA models using the above estimation and `auto.arima()`
- 5) Check residuals via `checkresiduals()` to see if any models are heavily flawed
- 6) Compare ARIMA models via AIC and a sanity check of forecasting trend and size of forecasting interval

After this process, `forecast()` was applied to all twenty-two models to generate ten-years forecasts for each model.

These forecasts were then rounded to the nearest integer, to match the nature of enrollment headcounts. Then, graduate and undergraduate predictions within each university were added together for a projection of total enrollment. Note that this step would ordinarily cause issues with confidence intervals, but the total projection is meant for visual inspection of the data, not as a robust model. Two final sets of projections were created, with one being the sum of all undergraduate, graduate, and total projections across all USM institutions, and the other being that same projection but excluding University of Maryland, Global Campus from the summation.

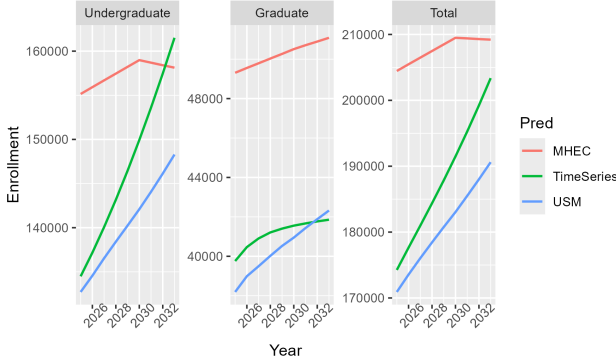
Finally, these forecasts were compared to professional models.

V. RESULTS

In order to save space and avoid copying others' results, Table I shows only the forecasts generated from the time series methods used in this paper. Thus, comparisons to professional forecasts will be done through closer examination of a few highlighted examples.

Firstly, comparing total enrollment gives a reasonable idea of the average performance of each model. The MHEC model produces a much more conservative estimate for the growth rate of undergraduate students, ranging from 155k at the initial enrollment in 2025 to a final prediction of 158k, with a slightly larger peak in 2030. By contrast, the USM model starts with 133k in 2025 and steadily increases to 148k in 2033. The time series model results in a much more dynamic prediction, with 134k enrolled in 2025 – much like the USM model – but a steady increase to 162k in 2033, which is notably higher than either of the other two models.

Enrollment Forecasts for University System of Maryland
2025 to 2033

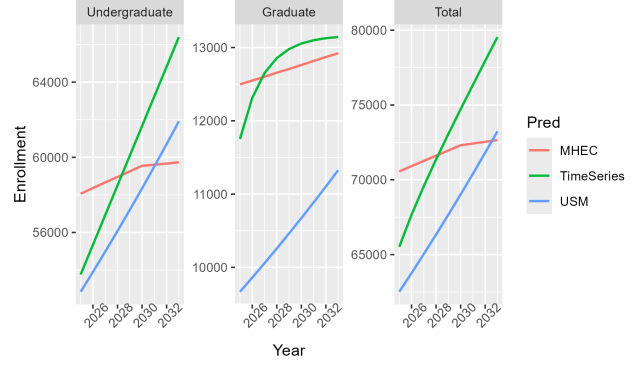


Repeating this comparison for graduate data, MHEC predicts a steady growth of enrollment from 49.3k to 51.1k. By contrast, both USM and the time series have much more conservative models, from 38.1k and 39.8k to 42.3k and 41.9k respectively.

Overall, the time series model matches and exceeds the growth rate of the USM model, but still does not quite approach the numbers predicted by the MHEC model.

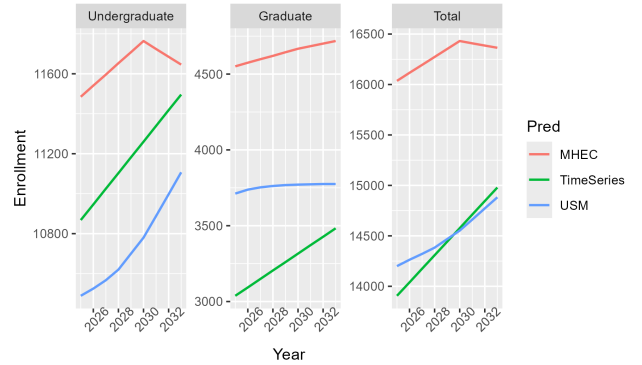
Next, consider individual universities. One university of major interest is the University of Maryland, Global Campus, due to its non-standard trend after 2000, and that it is the highest enrollment university within USM. For both of these reasons, predictive models for enrollment is worth highlighting. The MHEC model predicts a slow, steady growth of students, with undergraduate enrollment growing from 58k to 60k and graduate enrollment from 12.5k to 12.9k. The USM model predicts similar total enrollment in 2033, but has smaller early projections with higher growth. Specifically, 53k undergraduates become 62k, and 9.6k graduates become 11.3k, which is a significantly lower proportion of graduate students to undergraduates. This time series model predicts much higher numbers than both of these models again, with undergraduates going from 54k to 66k and graduates going from 11.8k to 13k. Given that UMGC is a significant proportion of total USM enrollment, the overprediction in this model may account for a large portion of the overall overprediction.

Enrollment Forecasts for University of Maryland, Global Campus
2025 to 2033



Another university worth highlighting is the University of Maryland, Baltimore County (UMBC), as the university's historical data has a similar trend to many other universities. Here, the MHEC model mimics the overall model, with high values, linear trends, and a peak in 2030. The USM model predicts an exponential increase, in contrast to its total enrollment appearing more linear. Finally, the time series model projects a linear increase in all categories.

Enrollment Forecasts for University of Maryland, Baltimore County
2025 to 2033



VI. CONCLUSION

This paper contributes a third model to the world of forecasting University System of Maryland enrollment rates. The disagreements between this model and the two existing professional forecasts further highlights the difficulties of giving accuracy forecasts for human behavior. However, this model consistently fits between both forecasts, implying that it is sufficient for most purposes and requires the least external data. More advanced techniques, different time series models, and adjustments for population or economic factors may help to refine this model for further accuracy. And, only time will tell as to which of these three models is most accurate.

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TABLE I
NINE-YEAR FORECAST FOR USM ENROLLMENT

University	Type	2025	2026	2027	2028	2029	2030	2031	2032	2033
Bowie	Undergrad	5009	4872	4740	4605	4472	4338	4204	4070	3936
	Graduate	1266	1290	1303	1309	1312	1314	1314	1315	1315
	Total	6275	6162	6043	5914	5784	5652	5518	5385	5251
Coppin	Undergrad	1933	1933	1933	1933	1933	1933	1933	1933	1933
	Graduate	303	303	303	303	303	303	303	303	303
	Total	2236	2236	2236	2236	2236	2236	2236	2236	2236
Frostburg	Undergrad	3462	3481	3487	3487	3482	3476	3468	3459	3450
	Graduate	682	682	682	682	682	682	682	682	682
	Total	4144	4163	4169	4169	4164	4158	4150	4141	4132
Salisbury	Undergrad	6246	6227	6197	6173	6146	6120	6094	6068	6042
	Graduate	737	737	737	737	737	737	737	737	737
	Total	6983	6964	6934	6910	6883	6857	6831	6805	6779
Towson	Undergrad	16133	16065	16030	16012	16003	15998	15996	15994	15994
	Graduate	3157	3207	3254	3321	3369	3424	3475	3531	3584
	Total	19290	19272	19284	19333	19372	19422	19471	19525	19578
UBalt	Undergrad	1813	2338	3068	3977	5078	6371	7851	9522	11383
	Graduate	1701	1647	1593	1539	1485	1431	1377	1323	1269
	Total	3514	3985	4661	5516	6563	7802	9228	10845	12652
UMB	Undergrad	960	960	960	960	960	960	960	960	960
	Graduate	5669	5665	5662	5661	5660	5660	5660	5660	5660
	Total	6629	6625	6622	6621	6620	6620	6620	6620	6620
UMBC	Undergrad	10868	10946	11025	11103	11182	11260	11339	11418	11496
	Graduate	3038	3093	3149	3205	3260	3316	3372	3427	3483
	Total	13906	14039	14174	14308	14442	14576	14711	14845	14979
UMD	Undergrad	31672	32211	32750	33289	33828	34367	34906	35445	35984
	Graduate	10744	10805	10831	10841	10845	10847	10847	10848	10848
	Total	42416	43016	43581	44130	44673	45214	45753	46293	46832
UMES	Undergrad	2630	2793	2956	3119	3281	3444	3607	3770	3933
	Graduate	711	726	741	757	772	787	802	817	832
	Total	3341	3519	3697	3876	4053	4231	4409	4587	4765
UMGC	Undergrad	53766	55346	56925	58504	60084	61663	63243	64822	66401
	Graduate	11752	12312	12651	12856	12979	13054	13099	13127	13143
	Total	65518	67658	69576	71360	73063	74717	76342	77949	79544
USM	Undergrad	134492	137172	140071	143162	146449	149930	153601	157461	161512
	Graduate	39760	40467	40906	41211	41404	41555	41668	41770	41856
	Total	174252	177639	180977	184373	187853	191485	195269	199231	203368
USM, no GC	Undergrad	80726	81826	83146	84658	86365	88267	90358	92639	95111
	Graduate	28008	28155	28255	28355	28425	28501	28569	28643	28713
	Total	108734	109981	111401	113013	114790	116768	118927	121282	123824

Abbreviations and shortenings for university names used to save space and preserve readability of this table.

APPENDIX LIST OF ARIMA MODELS USED

- Bowie State University:
 - Undergraduate - ARIMA(1,2,0)
 - Graduate - ARIMA(1,1,0)
- Coppin State University:
 - Undergraduate - ARIMA(0,1,1)
 - Graduate - ARIMA(0,1,0)
- Frostburg State University:
 - Undergraduate - ARIMA(1,2,1)
 - Graduate - ARIMA(0,1,0)
- Salisbury University:
 - Undergraduate - ARIMA(1,2,0)
 - Graduate - ARIMA(0,1,0)
- Towson University:
 - Undergraduate - ARIMA(1,1,0)
 - Graduate - ARIMA(3,2,0)
- University of Baltimore:
 - Undergraduate - ARIMA(2,3,0)
 - Graduate - ARIMA(0,2,0)
- University of Maryland, Global Campus:
 - Undergraduate - ARIMA(0,2,1)
 - Graduate - ARIMA(1,1,0)
- University of Maryland, Baltimore:
 - Undergraduate - ARIMA(0,1,0)
 - Graduate - ARIMA(1,1,0)
- University of Maryland, Baltimore County:
 - Undergraduate - ARIMA(0,2,1)
 - Graduate - ARIMA(0,1,1) with drift
- University of Maryland, College Park:
 - Undergraduate - ARIMA(0,2,0)
 - Graduate - ARIMA(1,1,0)
- University of Maryland, Eastern Shore:
 - Undergraduate - ARIMA(0,2,1)
 - Graduate - ARIMA(0,1,0) with drift

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